10.1 Sectors

Lesson Introduction

Benchmark	MA.912.GR.6.4 Solve mathematical and real-world problems involving the arc length and area of a sector in a given circle.	Learning Targets	• I can define a sector of a circle. • I can explain how the area of a sector is defined as a fractional part of the area of a circle.

Benchmark Coherence	Building On	Working Toward
	N/A	N/A

Essential Questions	• What is the relationship between the degree measure of a central angle and the degree measure of an entire circle? • How can you calculate the area of a sector when given a circle using ratios?
Lesson Narrative	Students will explore central angles and their measures in relation to the measure of the entire circle. Using this relationship, students will be able to find the area of a sector by using proportions.

Component Summary	10.1.1 Warm-Up (~5 min.)	Speculating about the game of Pac-Man *(SE p. 119)*	10.1.3 Guided Practice (10-15 min.)	Finding the area of sectors using proportions *(SE p. 121)*
	10.1.2 Exploration (10-15 min.)	Exploring fractions of a circle *(SE p. 120)*	10.1.4 Your Turn! (5-10 min.)	Calculating fractions, areas, and central angles of the sectors of a circle *(SE p. 123)*
	10.1.5 Wrap-Up (~5 min.)	Students summarize their learning *(SE p. 124)*		

Resources	Glossary	Materials	Additional Preparation
	Key Terms: • Central Angle Additional Glossary: • Sector of a Circle	• Video for Warm-Up 10.1.1 • (optional) Calculator	• Warm-Up 10.1.1 contains a video to accompany the question prompts.

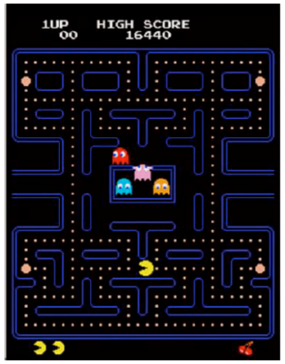
<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>• Students may have difficulty manipulating the expressions to isolate the sector area in the 10.1.3 Guided Practice.
• Students may use the formula of the circumference to calculate the area of a circle.
• Students may struggle with simplifying the ratio of the central angle to the whole circle, and not see the connection between the ratio and the sector fraction of the circle.</td><td>• Facilitate a conversation about the difference between finding the area of a circle and finding the circumference of a circle.
• Students should not be using the formal formula for calculating the area of a sector yet.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Clarify, Critique, Connect
Consider implementing this strategy by allowing students to trade their response in the 10.1.3 Guided Practice or the 10.1.4 Your Turn! with another student. Students will then ask their partner clarifying questions as needed. This process will allow students to think more analytically and provide a more detailed mathematical explanation.</td><td>MA.K12.MTR.1.1: Participate in Effortful Learning
Students will be working on finding the area of a sector by making connections between the fraction of a circle and the central angle. As students explore this connection, they need to ask clarifying questions to deepen their understanding. The 10.1.2 Exploration and the 10.1.3 Guided Practice provide help and support for students with the new concept. The 10.1.4 Your Turn! will provide students the confidence that they need to complete and work through complex and/or new problems.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">This lesson provides multiple visual entry points for students as they discover the relationship between the measure of the central angles, the area of the sector, and the area of the circle. Read and discuss problems aloud to assist students who benefit from auditory entry points.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

10.1.1 Warm-Up: Pac-Man

Component Overview: *During this Warm-Up, students will watch a short video (33 seconds) of Pac-Man going around the maze and eating dots. Students will make observations by writing down things they notice and wonder about Pac-Man's body as he opens and closes his mouth to eat dots.*

10.1.1 Warm-Up: Pac-Man

The classic and popular Pac-Man video game was released in Japan in May of 1980 and was in the United States by October of that same year. The yellow, pie-shaped character travels around a maze trying to eat dots while avoiding the ghosts hunting him.



Watch the video of Pac-Man going around the maze and eating dots.

1. What do you notice about Pac-Man's body as he opens and closes his mouth to eat the dots?
   Answers will vary. I noticed that his body is a whole circle when his mouth is closed and is a partial circle when his mouth is open. I noticed that his body get smaller when his mouth is opening and larger when it's closing.

2. What do you wonder as you watch Pac-Man move around the maze?
   Answers will vary. I wonder why he doesn't get larger as he consumes each pellet. I wonder what would cause Pac-Man to go faster throughout the maze. I wonder how many power pellets can fit inside Pac-Man and I wonder what those creatures are chasing him.

What do you notice about the location of Pac-Man's mouth in relation to his body? How is Pac-Man's body with an open mouth similar to a pizza missing one slice?

10.1.2 Exploration: Fractions of a Circle

Component Overview*: Students will be calculating the area of a given number of pizza slices. Then, students will be exploring the relationship between the fraction of a circle and its central angle in degrees. Students will have the opportunity to work with a partner to create a definition for a sector of a circle as it relates to a slice of pizza.*

10.1.2 Exploration: Fractions of a Circle

Work with your partner to complete the following.

1. Pizza sizes are given in terms of the pizza's diameter. The standard size for most medium circular pizzas is 14 inches (in.) and it is usually cut into 10 triangular slices.

   a. What is the approximate area of a 14-inch pizza? Use $\frac{22}{7}$ for π.

$A = \pi r^2$

$r = \frac{diameter}{2} = \frac{14}{2} = 7$

$A = \left(\frac{22}{7}\right)(7)^2$

$A = \left(\frac{22}{7}\right)49$

$A \approx 154\ in^2$

   b. If the slices are cut precisely and equally, what fraction and percentage of the pizza is 5 slices?

Fraction	Percentage
$\frac{5}{10} = \frac{1}{2}$	50%

   c. What is the area of 5 slices?

$\frac{154 in^2}{2} \approx 77\ in^2$

   d. What is the area of 1 slice?

$\frac{154 in^2}{10} \approx 15.4\ in^2$

Consider prompting questions to help students make sense of the problem as a check for understanding.

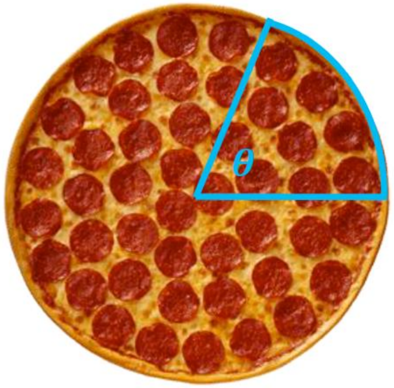
- *What is the radius of the circle?*
- *How can you find the area of a circle?*
- *What does it mean to use $\frac{22}{7}$ for π?*

Consider challenging students to find the area of 2 slices of pizza if the 14-inch pizza was cut into 8 triangular slices. Have students use $\frac{22}{7}$ for π. ($38.5\ in^2$)

10.1.2 Exploration: Fractions of a Circle (continued)

2. The point where a pizza is divided to create precisely equal slices is a central angle.

> A ***central angle*** is an angle with its vertex at the center of a circle with radii as its side.



The central angle is a fraction of the total $360°$ in a circle.

a. Complete the table. The first row of the table has been filled in.

	Fraction	Central Angle in Degrees (°)
Half of a Circle	$\frac{1}{2}$	$\left(\frac{1}{2}\right)\cdot(360°) = 180°$
Quarter of a Circle	$\frac{1}{4}$	$\left(\frac{1}{4}\right)\cdot(360°) = 90°$
Third of a Circle	$\frac{1}{3}$	$\left(\frac{1}{3}\right)\cdot(360°) = 120°$
Tenth of a Circle	$\frac{1}{10}$	$\left(\frac{1}{10}\right)\cdot(360°) = 36°$

Notice and Wonder
Consider providing students with the opportunity to write down things they notice and wonder about the central angle. This will allow students the chance to make observations to deepen their understanding of a central angle.

What relationship do you notice between the fraction of the circle and the central angle in degrees? How is this similar to finding the area of a given number of slices of pizza?

10.1.2 Exploration: Fractions of a Circle (continued)

b. Complete the statement.

One slice, or fraction of the circle, is bounded by two

- ○ diameters
- ○ radii

and an arc. Since each slice is the same size, the central angle for each fractional piece of the circular pizza will

- ○ vary.
- ○ be equal.

c. The pizza slices are an example of ***sectors***. With your partner, write a definition of the term ***sector***.
Answers may vary.
A sector is a fraction of the area of a circle.
A sector is a piece of a circle that is in the shape of a slice of pizza, with a maximum central angle of 360°.

d. Ask two other partner pairs to share their definition of a sector. Record a summary of their definition. *You are the only person who should write in the first column of the table.* Have the person initial next to your summary stating that your summary was correct.

My Summary of Their Definition of a Sector	Initials
Answers will vary. *A sector is a fraction of a circle with a central angle.*	
Answers will vary. *A sector is a piece of a circle, that looks like a slice of pizza, that represents a fraction of the circles' area.*	

For students who may struggle with showing their understanding of a sector, consider alternative ways to assess their learning and understanding such as recording themselves, or drawing and labeling a picture.

10.1.3 Guided Practice: Finding the Area of a Sector

Component Overview: *Students will build a connection between the sector fraction of a circle and the central angle of the sector. Students will have the opportunity to find the area of a given sector using the relationship between the sector fraction of a circle and the central angle of a sector.*

10.1.3 Guided Practice: Finding the Area of a Sector

The images below show a stopwatch with different time increments marked. The blue shading on the stopwatches each forms a sector.

1. Complete the table.

	Sector Fraction of the Circle	Central Angle Measure of the Sector	Ratio of the Central Angle to the Whole Circle
	$\frac{1}{12}$	$\left(\frac{1}{12}\right)\cdot(360°)=30°$	$\frac{30}{360}=\frac{1}{12}$
	$\frac{9}{12}$	$\left(\frac{9}{12}\right)\cdot(360°)=270°$	$\frac{270}{360}=\frac{3}{4}$
	$\frac{2}{12}$	$\left(\frac{2}{12}\right)\cdot(360°)=60°$	$\frac{60}{360}=\frac{1}{6}$

2. Summarize what you notice about the relationship between the values in each row.
   Answers may vary. The sector fraction of a circle is equal to the ratio of the central angle to the whole circle.

Think-Pair-Share
This strategy can be used on questions 1 and 2.

- *Students should work each problem individually and record answers.*
- *Students will pair with their partner or group members to discuss solutions and provide constructive critiques.*
- *The teacher will select a few groups to share their solutions and methods.*

Discussion provides additional entry points for auditory learners. Consider utilizing sentence stems, and provide ample time for students to grapple with academic language.

- *I think the central angle is _______ because the fraction of the circle is ______.*
- *I noticed that the sector fraction and the ratio of the central angle to the whole circle are________ which makes me think_____.*

10.1.3 Guided Practice: Finding the Area of a Sector (continued)

3. Discuss with your partner how ratios could be used to find the area of the sectors, including any additional information that might be needed. Summarize your discussion.

   Answers may vary.

   $$\frac{Central\ Angle}{360°} = \frac{Area\ of\ Sector}{Area\ of\ Circle}$$

   So, $Area\ of\ Sector = \frac{Area\ of\ Circle \cdot Central\ Angle}{360°}$

   In order to solve for the area of a sector, the area of the circle must be known so, additional information needed would be that of the radius.

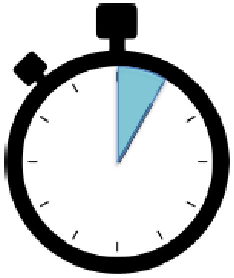
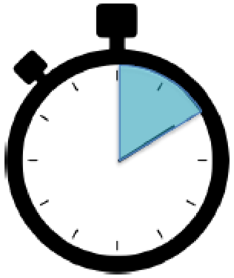
What does the area of a sector represent? How can proportions help you to find the area of a sector? What information will you need to find the area of a sector?

Given the radius of the stopwatch is 2 in., the area of the stopwatch is 4π in.^{2}.

$$A = \pi r^2$$
$$A = \pi(2)^2$$
$$A = 4\pi \text{ in.}^2$$

10.1.3 Guided Practice: Finding the Area of a Sector (continued)

4. Use the relationship between each sector and the whole stopwatch explored in problem 1 to find the area of each sector. Show your work.

	Area of the Blue-Shaded Sector
	$\frac{30}{360} = \frac{Sector\ Area}{4\pi}$ $Sector\ Area = \frac{30 \cdot 4\pi}{360}$ $Sector\ Area = \frac{120\pi}{360}$ $Sector\ Area = \frac{\pi}{3}\ in^2$
	$\frac{270}{360} = \frac{3}{4}$ $\frac{3}{4} = \frac{Sector\ Area}{4\pi}$ $Sector\ Area = \frac{3 \cdot 4\pi}{4}$ $Sector\ Area = 3\pi\ in^2$
	$\frac{60}{360} = \frac{1}{6}$ $\frac{1}{6} = \frac{Sector\ Area}{4\pi}$ $Sector\ Area = \frac{1 \cdot 4\pi}{6}$ $Sector\ Area = \frac{2\pi}{3}\ in^2$

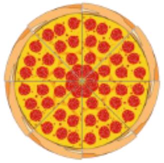
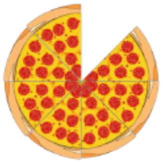
Emphasize that the sector area is a fraction of the area of the circle. Students should not be using the formal formula for calculating area of a sector yet.

10.1.4 Your Turn!

Component Overview: *Students will work individually to find the fraction of the circle, the area of the sector, and the central angle. Students will leave their answer for the area of the sector in terms of π.*

10.1.4 Your Turn!

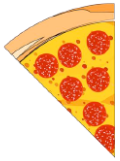
1. Kenya orders a large pizza that has a 16 in. diameter. Complete the table on the number of slices she has.

	Fraction of the Circle	Area of the Sector	Central Angle
Eight Slices 	$\frac{8}{8} = 1$	$\pi(8)^2$ $64\pi\ in^2$	$1 \cdot 360°$ $= 360°$
Seven Slices 	$\frac{7}{8}$	$\frac{7}{8} \cdot 64\pi$ $= 56\,\pi\ in^2$	$\frac{7}{8} \cdot 360°$ $= 315°$

Emphasize to students what it means to leave an answer in terms of π.

Consider challenging students who completed their work early to create a formula that can be used to find the area of a sector of any circle. Encourage students to include the central angle in the formula.

10.1.4 Your Turn! (continued)

One Slice 	$\frac{1}{8}$	$\frac{1}{8} \cdot 64\pi$ $= 8\pi\ in^2$	$\frac{1}{8} \cdot 360°$ $= 45°$

Consider using this time to explore more about what the students may still have questions about. This can be done in small groups, or by checking in on students.

10.1.5 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.

10.1.5 Wrap-Up: Cool Down

1. Write an explanation to a classmate who might have missed today's lesson, in your own words, about sectors.

   Answers will vary.